

Statistical Methods Midterm Examination

Question 1 (20 points)

Define and explain the following concepts:

- a) Maximum Likelihood Estimation (MLE)
- b) Method of Moments
- c) Asymptotic efficiency
- d) Cramér-Rao Lower Bound

Question 2 (25 points)

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with probability density function:

$$f(x; \theta) = \theta x^{\theta-1}, 0 < x < 1, \theta > 0$$

- a) Find the method of moments estimator for θ .
- b) Find the maximum likelihood estimator for θ .
- c) Show that the MLE is consistent.

Question 3 (25 points)

Consider testing $H_0: \mu = \mu_0$ versus $H_1: \mu \neq \mu_0$ for a normal distribution with unknown variance.

- a) Derive the likelihood ratio test statistic.
- b) Show that this test is equivalent to a two-tailed t-test.
- c) Discuss the relationship between confidence intervals and hypothesis tests in this context.

Question 4 (20 points)

Explain the concept of sufficiency and provide an example. State and use the Factorization Theorem to show that a statistic is sufficient for a parameter.

Question 5 (10 points)

What is the difference between a consistent estimator and an unbiased estimator? Can an estimator be consistent but biased? Provide an example.

Formula Sheet Reference:

- Likelihood Function: $L(\theta) = \prod_i f(x_i; \theta)$
- MLE: θ maximizes $L(\theta)$ or $\log L(\theta)$
- Cramér-Rao Bound: $\text{Var}(\theta) \geq 1/I(\theta)$, where $I(\theta)$ is Fisher Information
- Factorization Theorem: T is sufficient if $f(x; \theta) = g(T(x); \theta)h(x)$